School of Information Technologies and Engineering, ADA University

CSCI3613 – Artificial Intelligence

Fall 2025

Course Project Report

Team <3>

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Project

AI-Based Classroom and Course Scheduling System for School of IT and Engineering

Problem formulation

Formulate the problem. What type of task is it? (50-100 words)

The aim of this project is to address the university course timetabling problem, which involves assigning a set of course sections to available time slots, rooms, and instructors while satisfying a predefined set of constraints. Each course section consists of two weekly meetings that must be scheduled on different days. The primary hard constraints include instructor availability, avoidance of time and resource conflicts, room capacity limitations, consistency of instructor assignment across meetings of the same section, and additional institutional rules such as limits on daily master-level courses and restrictions on the number of instructors assigned to multi-section courses.

The problem is formally modeled as a **Constraint Satisfaction Problem (CSP)**. Each meeting of a course section is treated as a variable, whose domain consists of feasible combinations of day, time slot, room, and instructor. Hard constraints restrict illegal assignments, while soft constraints capture preferences such as limits on daily and weekly instructor workload. The objective is to find a complete assignment that satisfies all hard constraints while minimizing the total penalty induced by soft constraint violations. Due to the combinatorial nature of the problem and the exponential growth of the search space, efficient search strategies and heuristic guidance are essential.

EDA and data preprocessing

Describe your EDA and data preprocessing steps. Justify your choices. Include any figures and graphs.

The dataset used in this project was provided as a multi-sheet Excel file containing course information, instructor–course assignments, room capacities, and instructor availability constraints. During the exploratory data analysis (EDA) phase, the distribution of course sections across instructors, rooms, and time slots was examined to identify structural imbalances and potential scheduling bottlenecks. Preprocessing involved validating the uniqueness of section identifiers, handling missing or inconsistent entries, standardizing time representations into numeric formats, and converting raw spreadsheet data into structured Python data types such as dictionaries and lists. Feasible room assignments were derived by comparing room capacities with enrollment sizes. Instructor availability was processed into explicit time-interval constraints. Additionally, instructor workload distributions were analyzed to support later

evaluation of fairness and to motivate the inclusion of soft workload constraints in the optimization process

Room ID	Capacity (seats)
A102	45
A109	55
B101	40
B201	35
D108	30
D208	40
D216	20
D217	20
D015	15
D016	15
D017	15
B001	25
B002	25
B003	25

InstructorID	Day	StartTime	EndTime	Availability Type
INST001	Monday	13:00	17:15	UNAVAILABLE
INST001	Tuesday	13:00	17:15	UNAVAILABLE
INST001	Wednesday	13:00	17:15	UNAVAILABLE
INST001	Thursday	8:30	11:15	UNAVAILABLE
INST001	Friday	8:30	11:15	UNAVAILABLE
INST002	Monday	8:30	11:15	UNAVAILABLE
INST002	Tuesday	8:30	11:15	UNAVAILABLE
INST002	Wednesday	8:30	11:15	UNAVAILABLE
INST002	Thursday	8:30	11:15	UNAVAILABLE
INST002	Friday	8:30	11:15	UNAVAILABLE
INST003	Monday	13:00	17:15	UNAVAILABLE
INST003	Tuesday	8:30	11:15	UNAVAILABLE

InstructorID	Instructor Name	Course Name	COURSE ID
PROF001	Abzettin Adamov	Artificial Intelligence	CSCI6511
PROF001	Abzettin Adamov	Applied Machine Learning & Data Analytics	CSCI6767
PROF002	Jamaladdin Hasanov	Intro. to Computer Vision	CSCI6527
PROF002	Jamaladdin Hasanov	Database Systems	CSCI3615
PROF002	Jamaladdin Hasanov	Business Process Modeling for IT Solutions	INFT3610
PROF002	Jamaladdin Hasanov	Production & Operations Management	INFT3611
PROF003	Samir Rustamov	Natural Language Processing	CSCI6515
PROF003	Samir Rustamov	Artificial Intelligence	CSCI6511
PROF005	Araz Yusubov	Game Development Fundamentals	CSCI4836

COURSE ID	COURSE	SECTION ID	# OF STUDENTS
CSCI1202	Programming Principles II	20308	35
CSCI1202	Programming Principles II	20309	35
CSCI2303	Intro to Computer Networks	20317	30
CSCI2303	Intro to Computer Networks	20613	35
CSCI2304	Data Structures & Algorithms	20336	40
CSCI2304	Data Structures & Algorithms	20337	40
CSCI2407	Theory of Computation	20323	35
CSCI2407	Theory of Computation	20324	35

Implementation

Describe your implementation steps. Which algorithm(s) did you apply and why? What design decisions did you make? How did you evaluate your algorithm? Insert any figures.

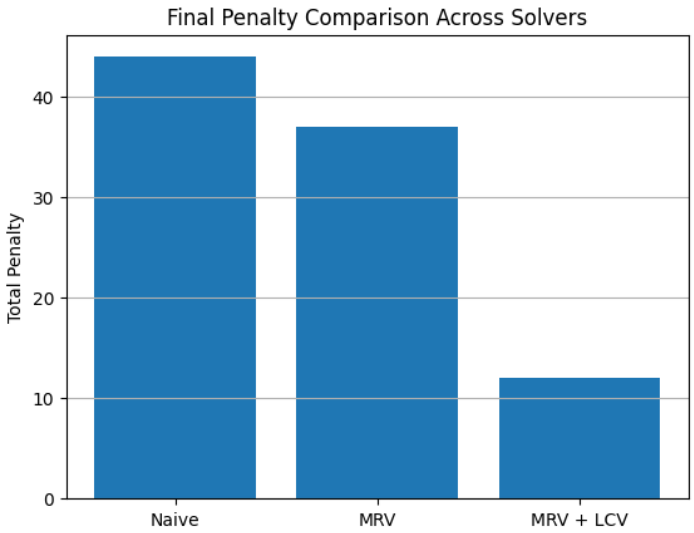
The solution was implemented in Python using a CSP-based backtracking framework. Each course section was decomposed into two meeting variables, significantly increasing modeling realism while also enlarging the search space. The domain of each variable consisted of all feasible combinations of day, time slot, room, and instructor that satisfied basic availability and capacity constraints.

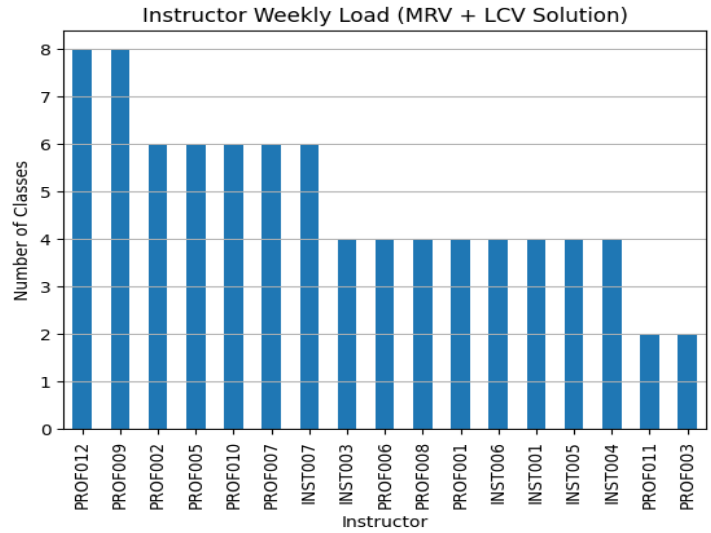
A branch-and-bound backtracking algorithm was employed to explore the solution space, with early pruning based on accumulated soft-constraint penalties. Several constraint-checking functions were implemented to enforce institutional rules, including instructor consistency across meetings, avoidance of time and resource conflicts, limits on daily master-level meetings, and restrictions on instructor usage across multi-section courses.

To improve efficiency, heuristic strategies were incorporated. The Minimum Remaining Values (MRV) heuristic was used to dynamically select the next variable with the smallest feasible domain, while the Least Constraining Value (LCV) heuristic ordered candidate assignments to minimize future conflicts. The solver was instrumented to collect detailed search-tree statistics, including nodes expanded, maximum search depth, and branching factors, enabling quantitative comparison of different heuristic configurations.

Experiments

What experiments did you run? Insert any figures.





Several experiments were conducted to evaluate the impact of heuristic strategies on search efficiency and solution quality. A baseline solver using fixed variable ordering and branch-and-bound backtracking was first executed. This was followed by heuristic-enhanced solvers incorporating MRV alone and a combination of MRV and LCV.

For each configuration, execution time, total penalty, and search-tree statistics were recorded. In particular, the number of expanded nodes, maximum depth reached, and average branching factor were analyzed to characterize the structure of the search process. Additionally, instructor workload distributions were visualized to assess the practical quality of the generated timetables. These experiments enabled a systematic comparison between uninformed and heuristic-guided search strategies.

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⚠ Total penalty = 44
   (Daily excess weight=5, Weekly excess weight=3)

===== FINAL SCHEDULE =====

Section ID  Meeting Course ID                                  Course       Day Start Time Room Instructor
     20308 20308_M1  CSCI1202                   Programming Principles II    Monday      08:30 A102    INST005
     20309 20309_M1  CSCI1202                   Programming Principles II    Monday      08:30 A109    INST001
     20317 20317_M1  CSCI2303                  Intro to Computer Networks    Monday      08:30 B101    PROF012
     20336 20336_M1  CSCI2304                Data Structures & Algorithms    Monday      08:30 D208    PROF009
     20323 20323_M1  CSCI2407                       Theory of Computation    Monday      08:30 B201    INST003
     20365 20365_M1  MATH1222                                  Calculus II    Monday      08:30 D108    PROF011
     20613 20613_M1  CSCI2303                  Intro to Computer Networks    Monday      10:00 A102    PROF012
     20337 20337_M1  CSCI2304                Data Structures & Algorithms    Monday      10:00 A109    PROF009
     20324 20324_M1  CSCI2407                       Theory of Computation    Monday      10:00 B101    INST003
     20319 20319_M1  CSCI3510               Princpls of Operating Systems    Monday      10:00 B201    INST004
     20296 20296_M1  CSCI3612               Object Oriented Analy.&Design    Monday      10:00 D208    PROF005
     20295 20295_M1  CSCI4701                                Deep Learning    Monday      10:00 D108    INST001
     20320 20320_M1  CSCI3510               Princpls of Operating Systems    Monday      11:30 A102    INST004
     20595 20595_M1  CSCI3600                 Intro to Python Programming    Monday      11:30 A109    PROF009
     20297 20297_M1  CSCI3612               Object Oriented Analy.&Design    Monday      11:30 B101    PROF005
     20328 20328_M1  INFT2405                            Web & Mobile II    Monday      11:30 B201    INST005
     20301 20301_M1  INFT3610 Business Process Modeling for IT Solutions    Monday      11:30 D208    PROF008
     20366 20366_M1  MATH1222                                  Calculus II    Monday      11:30 D108    PROF011
     20596 20596_M1  CSCI3600                 Intro to Python Programming    Monday      13:00 A102    PROF009
     20321 20321_M1  CSCI3615                           Database Systems    Monday      13:00 A109    PROF012
     20302 20302_M1  INFT3610 Business Process Modeling for IT Solutions    Monday      13:00 B101    PROF008
     20546 20546_M1  MATH3501                               Linear Algebra    Monday      13:00 D208    PROF007
     20568 20568_M1  SITE1101                 Principles of Info. Systems    Monday      13:00 B201    PROF010
     20294 20294_M1  CSCI4700                                  Data Mining    Monday      14:30 A102    INST006
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⚠ Total penalty = 12

===== SEARCH TREE STATISTICS =====
Nodes expanded: 8398
Max depth reached: 82 (theoretical max = 82)
Average branching factor: 86.11
Maximum branching factor: 254

===== FINAL SCHEDULE =====

Section ID  Meeting Course ID                                  Course       Day Start Time Room Instructor
     20613 20613_M2  CSCI2303                 Intro to Computer Networks    Monday      08:30 B201    PROF012
     20294 20294_M1  CSCI4700                                Data Mining    Monday      08:30 A102    INST003
     20317 20317_M1  CSCI2303                 Intro to Computer Networks    Monday      10:00 D108    PROF012
     20595 20595_M2  CSCI3600                Intro to Python Programming    Monday      10:00 B201    PROF009
     20596 20596_M1  CSCI3600                Intro to Python Programming    Monday      13:00 B201    PROF009
     20365 20365_M1  MATH1222                                Calculus II    Monday      13:00 A102    PROF006
     20366 20366_M1  MATH1222                                Calculus II    Monday      13:00 A109    PROF011
     20568 20568_M1  SITE1101                Principles of Info. Systems    Monday      13:00 D108    PROF010
     20296 20296_M2  CSCI3612              Object Oriented Analy.&Design    Monday      16:00 B201    INST004
     20297 20297_M2  CSCI3612              Object Oriented Analy.&Design    Monday      16:00 D208    PROF005
     20541 20541_M1  CSCI6527                  Intro. to Computer Vision    Monday      16:00 D216    PROF002
     20542 20542_M1  CSCI6767  Applied Machine Learning & Data Analytics    Monday      16:00 B001    PROF001
     20328 20328_M2  INFT2405                            Web & Mobile II    Monday      16:00 A102    INST007
     20546 20546_M2  MATH3501                             Linear Algebra    Monday      16:00 A109    PROF007
     20547 20547_M1  MATH3501                             Linear Algebra    Monday      16:00 B101    PROF006
     20323 20323_M1  CSCI2407                      Theory of Computation    Monday      17:30 B201    PROF010
     20330 20330_M2  INFT2303                  Systems Analysis & Design    Monday      17:30 B101    INST006
     20355 20355_M2  INFT2404              IT Systems: Hardware&Software    Monday      17:30 D208    INST007
     20302 20302_M2  INFT3610 Business Process Modeling for IT Solutions    Monday      17:30 A102    PROF002
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⚠ Total penalty = 37

===== SEARCH TREE STATISTICS (MRV ONLY) =====
Nodes expanded: 8062
Max depth reached: 82 (theoretical max = 82)
Average branching factor: 82.19
Maximum branching factor: 249

===== FINAL SCHEDULE =====

Section ID  Meeting Course ID                                  Course       Day Start Time Room Instructor
     20336 20336_M1  CSCI2304               Data Structures & Algorithms    Monday      08:30 B101    PROF009
     20319 20319_M1  CSCI3510              Princpls of Operating Systems    Monday      08:30 B201    INST004
     20321 20321_M1  CSCI3615                          Database Systems    Monday      08:30 A109    PROF012
     20294 20294_M1  CSCI4700                                Data Mining    Monday      08:30 B001    INST003
     20295 20295_M1  CSCI4701                              Deep Learning    Monday      08:30 D108    INST001
     20619 20619_M1  CSCI4836              Game Development Fundamentals    Monday      08:30 A102    PROF005
     20301 20301_M1  INFT3610 Business Process Modeling for IT Solutions    Monday      08:30 D208    PROF008
     20337 20337_M1  CSCI2304               Data Structures & Algorithms    Monday      10:00 A109    PROF009
     20320 20320_M1  CSCI3510              Princpls of Operating Systems    Monday      10:00 B201    INST004
     20322 20322_M1  CSCI3615                          Database Systems    Monday      10:00 A102    PROF012
     20302 20302_M1  INFT3610 Business Process Modeling for IT Solutions    Monday      10:00 B101    PROF008
     20365 20365_M1  MATH1222                                Calculus II    Monday      10:00 D108    PROF006
     20359 20359_M1  MATH3601                        Functional Analysis    Monday      10:00 D208    PROF011
     20308 20308_M1  CSCI1202                  Programming Principles II    Monday      11:30 B201    INST001
     20613 20613_M1  CSCI2303                 Intro to Computer Networks    Monday      11:30 A102    PROF012
     20595 20595_M1  CSCI3600                Intro to Python Programming    Monday      11:30 A109    PROF009
     20328 20328_M1  INFT2405                            Web & Mobile II    Monday      11:30 B101    INST005
     20366 20366_M1  MATH1222                                Calculus II    Monday      11:30 D108    PROF011
     20286 20286_M1  MATH3601                        Functional Analysis    Monday      11:30 D208    PROF007
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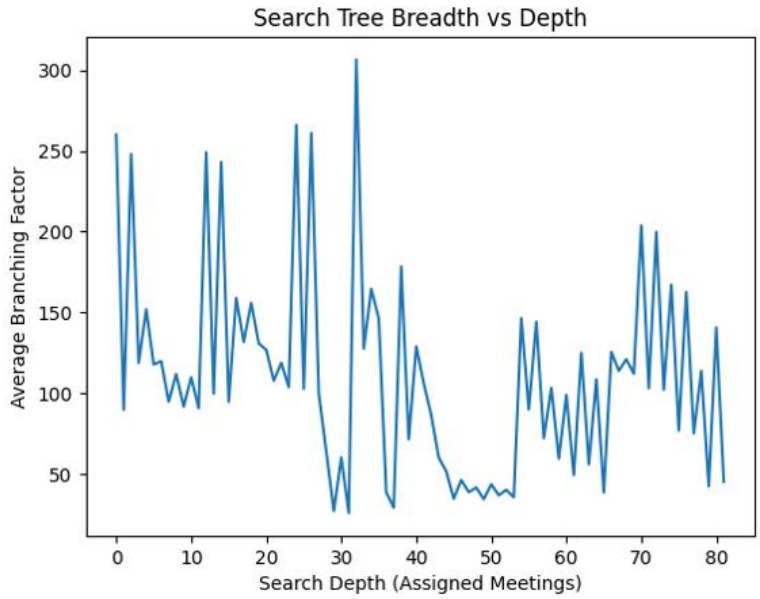
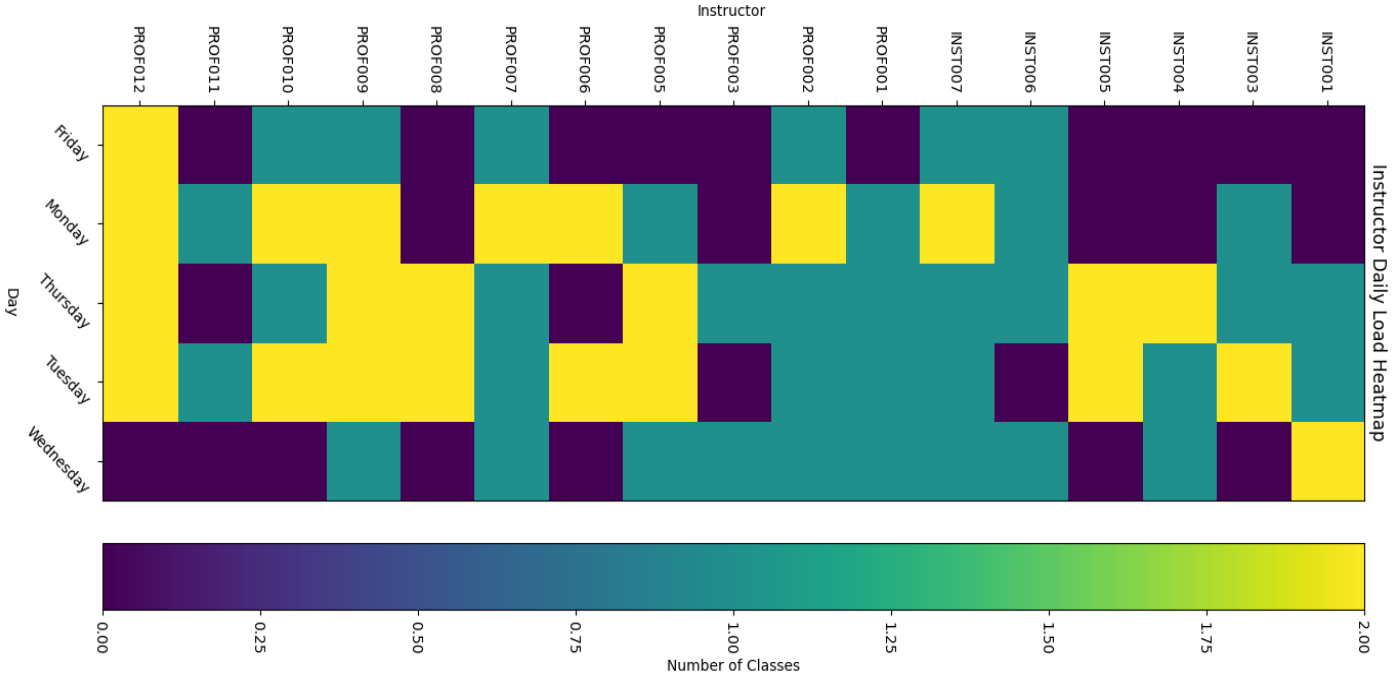
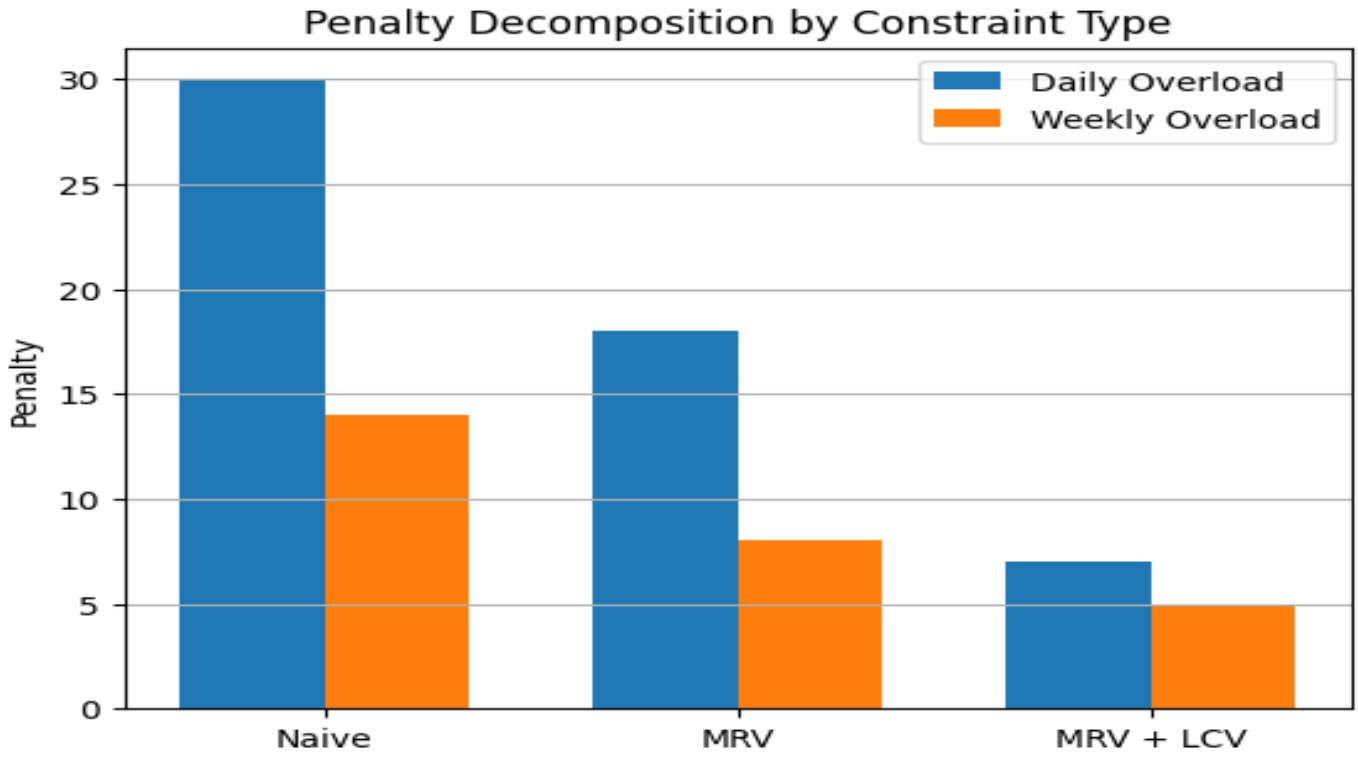

Search Tree Breadth vs Depth
Average Branching Factor
Search Depth (Assigned Meetings)
300
250
200
150
100
50
0
10
20
30
40
50
60
70
80

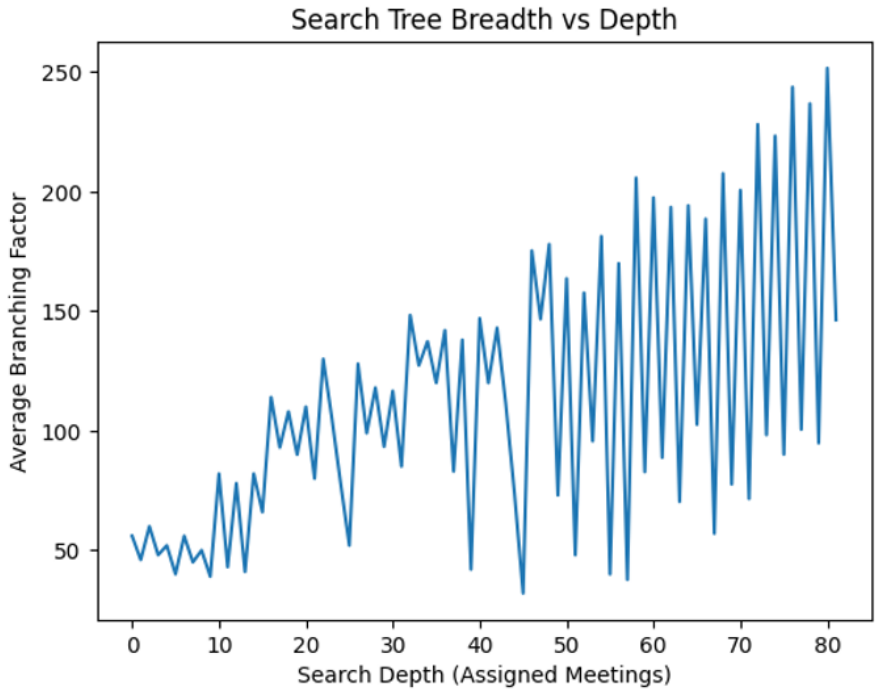

Instructor Daily Load Heatmap
Instructor
Day
Friday
Monday
Thursday
Tuesday
Wednesday
PROF012
PROF011
PROF010
PROF009
PROF008
PROF007
PROF006
PROF005
PROF003
PROF002
PROF001
INST007
INST006
INST005
INST004
INST003
INST001
Number of Classes
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00



Discussion of results

How do you interpret the results of the project? Discuss the key points. (100-200 words)

The experimental results demonstrate that heuristic-guided CSP methods substantially improve the efficiency of the course timetabling process. Compared to the fixed-order baseline, the use of MRV significantly reduced the effective branching factor by prioritizing the most constrained variables, while the addition of LCV further improved performance by selecting values that preserved future flexibility.

All generated schedules satisfied the full set of hard constraints, and soft constraint penalties were notably reduced in heuristic-enhanced runs. Search-tree statistics confirmed a marked reduction in explored nodes and improved pruning behavior. Furthermore, the resulting timetables exhibited balanced instructor workloads, supporting their practical applicability. These findings indicate that the proposed CSP formulation, combined with MRV and LCV heuristics, provides a robust and scalable approach to university course timetabling. Future work may extend the model to incorporate instructor preferences, additional soft constraints, and larger institutional datasets.

This plot shows the average branching factor as the search progresses. The non-monotonic behavior is expected because we use MRV, which dynamically selects the most constrained variable at each step. As constraints accumulate unevenly, some depths have many options while others have very few. Despite the oscillations, the solver reaches full depth while expanding only 8,398 nodes, showing that the heuristics significantly reduce the effective search space.
Shouldn't this decrease? That would be true with a fixed variable order. With MRV, depth corresponds to the order in which constrained variables are resolved, so branching naturally fluctuates.